

# Sprint 4 Report

## CSE 4316

Charles Babbage

December 1st, 2026

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| Sprint start date: | November 17th, 2026                                             |
| Sprint end date:   | December 1st, 2026                                              |
| Project title:     | AquaGuard Smart Water-Leak Monitor                              |
| Team name:         | Team Nimbus                                                     |
| Partners:          | Alan Turing<br>Grace Hopper<br>John Von Neumann<br>Ada Lovelace |
| Instructor:        | Christopher D. McMurrough                                       |

## 1 Sprint Goal

Achieve reliable BLE pairing and a live sensor read from the device in the app on both iOS and Android, ahead of the December milestone demo.

## 2 Sprint Backlog

Planned tasks (hours).

| Task description                                      | Estimated work | Actual work |
|-------------------------------------------------------|----------------|-------------|
| Define GATT profile (services/characteristics)        | 4.0            | 4.0         |
| Firmware: advertise + reading/battery characteristics | 6.0            | 7.0         |
| App: scan, pair/bond, subscribe (iOS)                 | 5.0            | 6.0         |
| App: same flow on Android + parity test               | 5.0            | 5.5         |
| Bonded/encrypted link verification (SE-01)            | 3.0            | 3.0         |
| Power sampling to seed the battery budget             | 3.0            | 3.0         |
| <b>TOTAL</b>                                          | <b>26.0</b>    | <b>28.5</b> |

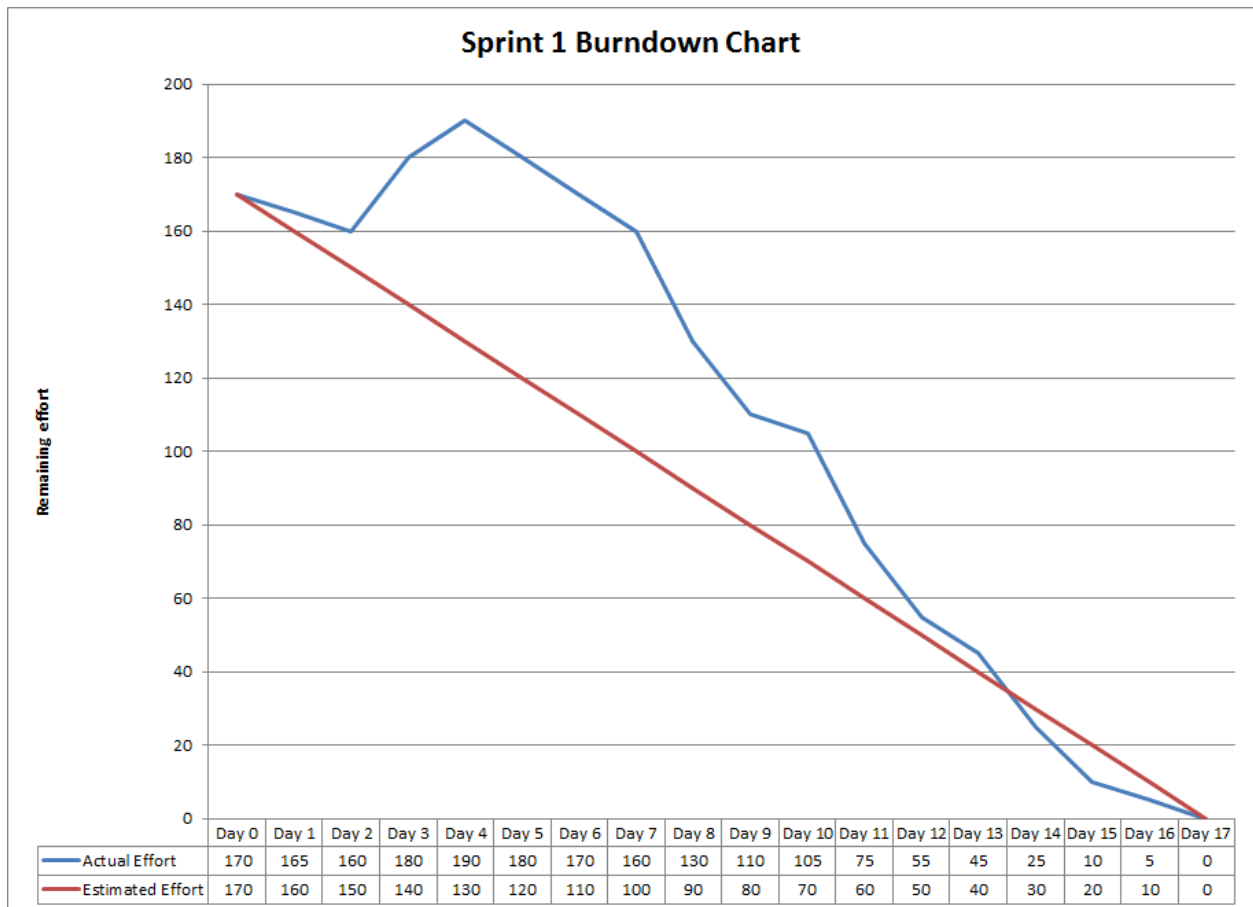
### 3 Individual Time Expenditures

Tasks performed by the individual (hours).

| Task Description                      | Actual work | Percent complete |
|---------------------------------------|-------------|------------------|
| Define GATT profile                   | 4.0         | 100%             |
| Firmware: advertise + characteristics | 7.0         | 100%             |
| Power sampling for battery budget     | 3.0         | 80%              |
| <b>TOTAL</b>                          | <b>14.0</b> | <b>-</b>         |

### 4 Team Burndown Chart

Ideal and actual daily team effort are shown below.



## 5 Individual Retrospective

Actions to **start** doing (top 3)

- Test BLE background behavior on real iOS/Android devices each sprint, not at the end.
- Record power measurements in a shared sheet to track the battery budget over time.
- Freeze the GATT profile as a versioned interface doc before app work.

Actions to **stop** doing (top 3)

- Testing BLE only on the simulator/emulator.
- Changing characteristic UUIDs without bumping the profile version.
- Leaving the radio always-on during bench tests (skews battery numbers).

Actions to **continue** doing (top 3)

- Pairing with John on the BLE central/peripheral contract.
- Keeping platform BLE differences isolated in one module (QA-01).
- Short daily blocker syncs.

## 6 Peer Review

Candid individual assessment; not shared. Each item graded 1–5.  
Evaluation items:

- **Participation** - present and contributing?
- **Communication** - timely, effective updates?
- **Work quality** - claimed tasks; met objectives; added value?
- **Professionalism** - professional and collaborative?
- **Overall** - overall value this period.

| Team member name | Participation | Communication | Work quality | Professionalism | Overall |
|------------------|---------------|---------------|--------------|-----------------|---------|
| Charles Babbage  | 5             | 5             | 5            | 5               | 5       |
| Alan Turing      | 5             | 4             | 5            | 5               | 5       |
| Grace Hopper     | 4             | 5             | 4            | 5               | 4       |
| John Von Neumann | 5             | 5             | 5            | 4               | 5       |
| Ada Lovelace     | 4             | 4             | 4            | 5               | 4       |